

diff - diff

Callaway-Sant'Anna. Sun-Abraham.

Borusyak-Jaravel-Spiess. Stacked DiD.

**One regression
to rule them all.**

LP-DiD: Local Projections DiD.

Now in diff-diff.

Dube, Girardi, Jordà & Taylor (2025).

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The staggered revolution left you with a shelf.

Callaway-Sant'Anna

Group-time ATTs.

Its own aggregation layer.

Sun-Abraham

Interaction-weighted ES.

Its own saturated model.

Borusyak-Jaravel-Spiess

Imputation.

Its own two-step machinery.

Stacked DiD

Dataset duplication.

Its own weighting scheme.

Every fix for TWFE's negative weights brought its own framework.

Same target. Four mental models.

Same question.

Four different machines.

Staggered State Policy

Minimum wage rolls out state by state. CS? Stacked? Which aggregation?

Phased Product Rollout

The feature ships to markets in waves. Sun-Abraham? Which saturation?

Programs That Switch Off

The retention treatment ends. Absorbing-only estimators don't apply.

Survey-Weighted Panels

CPS-style pweights, strata, PSUs. Which estimator takes a design?

LP-DiD.

It's just a regression.

A local projection per horizon: long difference,
time fixed effects, clean controls. Plain OLS.

One long difference per horizon.

OLS on clean controls. You can read every comparison it makes.

Weights provably non-negative.

The TWFE negative-weighting pathology cannot come back (Eqs. 9-10).

The estimand is a dial.

Variance-weighted for precision, or equal-weighted - your call.

Local projections (Jordà 2005), pointed at DiD.

Clean controls, horizon by horizon.



Each horizon is one regression you could run by hand.

Already-treated units never serve as controls.

The zoo was one regression all along.

$$y_{i,t+h} - y_{i,t-1} = \beta_h \Delta D_{it} + \delta_t^h + e_{it}^h$$

$$E(\beta_h) = \sum_g \omega_{g,h} \tau_h^g, \quad \omega_{g,h} \geq 0$$

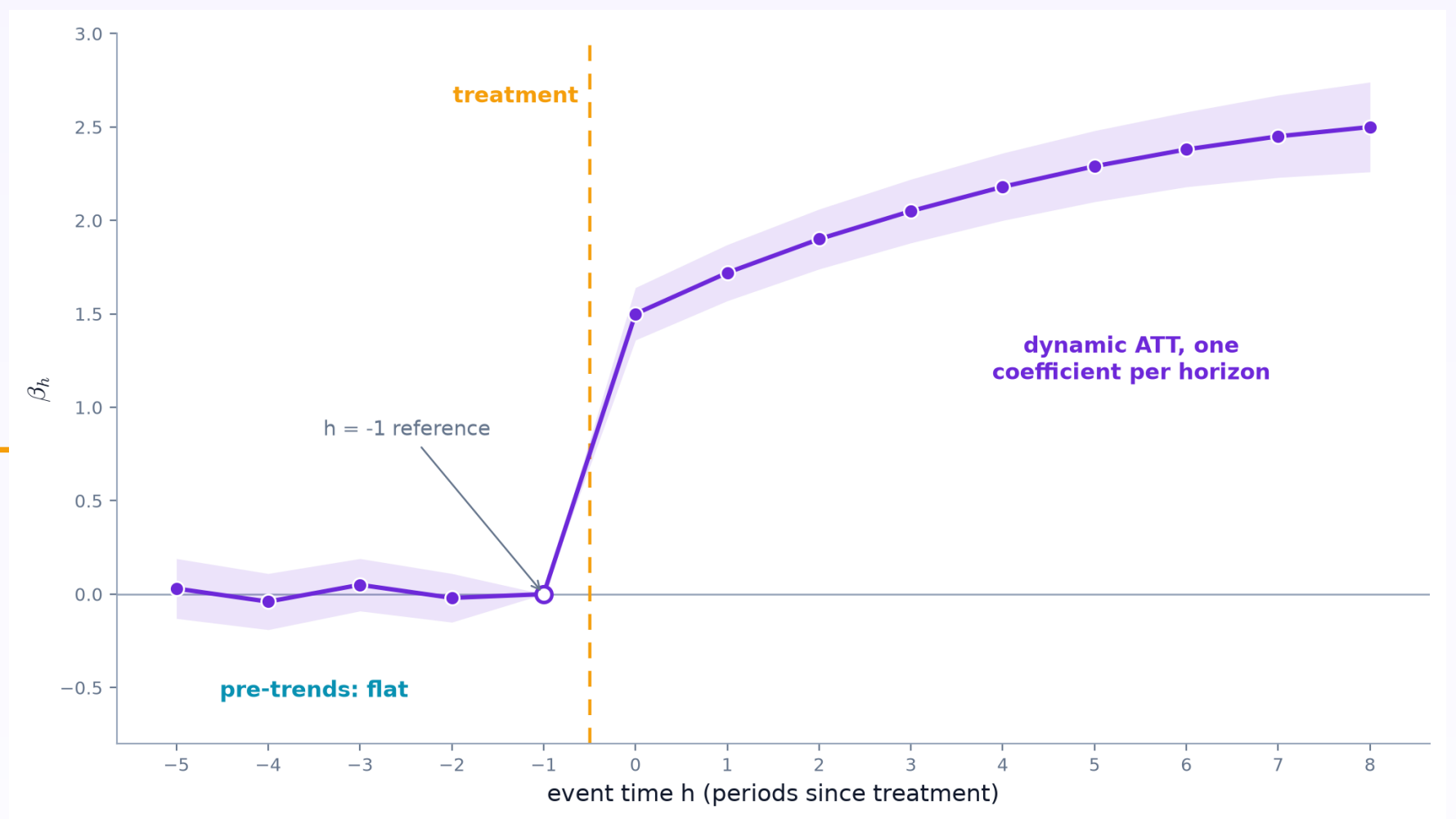
always non-negative -- no forbidden comparisons



<code>reweight=True</code>	== Callaway-Sant'Anna	<i>exact (Sec. 3.7)</i>
<code>default (variance-weighted)</code>	== Cengiz et al. stacked	<i>exact (Sec. 3.7)</i>
<code>h=0, single cohort</code>	== classic 2x2 DiD	<i>exact (Sec. 2.2)</i>
<code>pmd= 'max '</code>	-> BJS imputation	<i>exact single-cohort (fn. 10-11)</i>

Numerical equivalences proven in the paper - not analogies.

Pre-trends are just negative horizons.



The event study is native - not a post-hoc aggregation.

The Code.

Same sklearn-like API as every diff-diff estimator.

```
from diff_diff import LPDiD

result = LPDiD(
    pre_window=5, post_window=10,
).fit(
    data,
    outcome='revenue', unit='store',
    time='week', treatment='treated',
)

print(result.att)          # pooled post ATT = 2.31
result.event_study         # one beta per horizon

# flip one switch:
#   reweight=True   -> Callaway-Sant'Anna
#   non_absorbing=  -> treatments that switch off
#   survey_design=  -> pweights + strata + PSUs
```

The whole zoo is a keyword argument.

Production-ready.

Cluster-Robust SEs

Unit-level by default,
matches Stata `lstdid`

Covariates, Two Ways

RA path (BJS-style) or
direct with guardrails

PMD Baselines

Premean differencing +
pooled pre/post ATTs

Non-Absorbing Treatment

First-entry & effect-
stabilization (Eq. 12/13)

Survey Designs

pweights, strata, PSU,
FPC via Binder TSL

Composition Control

`no_composition` fixes the
sample across horizons

All composable on the basic `fit()` call.

Validated.

Against the authors' own tooling - and against our own shelf.

Authors' R recipes (danielegirardi/lpdid)

Event-study and pooled estimands match to $\sim 1e-13$.

Independent fixest reconstruction

Non-absorbing Eq. 12/13: point + SE parity to $\sim 1e-13$.

survey::svyglm, end to end

Survey path: per-horizon point, SE, and df all pinned.

Equivalences tested, not cited

reweight == our CallawaySantAnna; PMD == our ImputationDiD (1 cohort).

Now in diff-diff.

LP-DiD.

```
$ pip install --upgrade diff-diff
```

github.com/igerber/diff-diff

diff - diff

Difference-in-Differences for Python